# Lecture 4: Naïve Bayes

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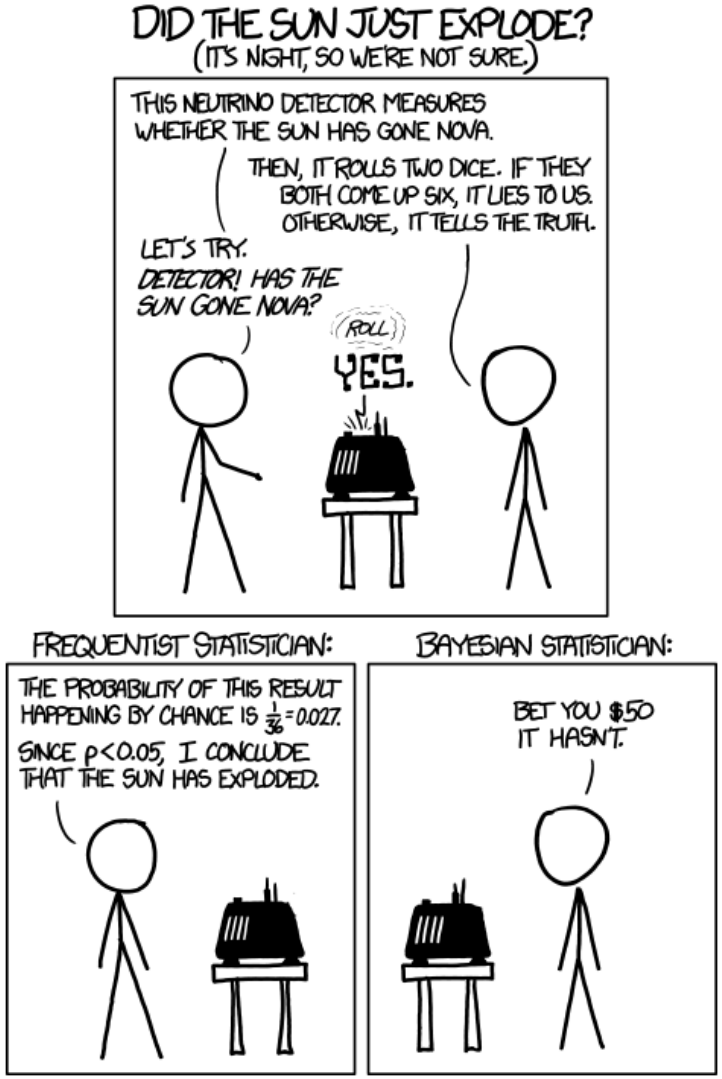


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# Naïve Bayes

- minimum probability of error using Bayes' rule
- naïve Bayes
- unigrams and bigrams
- estimating the likelihood: maximum likelihood parameter estimation
- Laplace smoothing

# MPE = MAP using Bayes' rule

$$f(x) = \underset{y}{\operatorname{argmax}}\, P(Y = y | X = x)$$

$$= \underset{y}{\operatorname{argmax}} \frac{P(Y = y)P(X = x | Y = y)}{P(X = x)}$$

$$= \underset{y}{\operatorname{argmax}}\, P(Y = y)P(X = x | Y = y)$$

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# The problem with likelihood: Too many words

What does it mean to say that the words, x, have a particular probability?

Suppose our training corpus contains two sample emails:

Email1: $Y = \text{spam}, X =$"Hi there man – feel the vitality! Nice meeting you…"

Email2: $Y = \text{ham}, X =$"This needs to be in production by early afternoon…"

Our test corpus is just one email:

Email1: *X*="Hi! You can receive within days an approved prescription for increased vitality and stamina"

How can we estimate $P(X =$ "Hi! You can receive within days an approved prescription for increased vitality and stamina"$|Y = \text{spam})$?

# Naïve Bayes: the "Bag-of-words" model

We can estimate the likelihood of an e-mail by pretending that the e-mail is just a bag of words (order doesn't matter).

With only a few thousand spam e-mails, we can get a pretty good estimate of these things:

- $P(W = \text{“hi”} | Y = \text{spam}), P(W = \text{“hi”} | Y = \text{ham})$
- $P(W = \text{“vitality”} | Y = \text{spam}), P(W = \text{“vitality”} | Y = \text{ham})$
- $P(W = \text{“production”} | Y = \text{spam}), P(W = \text{“production”} | Y = \text{ham})$

Then we can approximate $P(X|Y)$ by assuming that the words, $W$, are ***conditionally independent of one another given the category label***:

$$P(X = x | Y = y) \approx \prod_{i=1}^{n} P(W = w_i | Y = y)$$

# Naïve Bayes Representation

- Goal: estimate likelihoods P(Document | Class) and priors P(Class)
- Likelihood: ***bag of words*** representation
  - The document is a sequence of words $[w_1, w_2, \ldots, w_n]$
  - The order of the words in the document is not important
  - Each word is conditionally independent of the others given document class

Dear Sir.

First, I must solicit your confidence in this transaction, this is by virture of its nature as being utterly confidencial and top secret. …

TO BE REMOVED FROM FUTURE MAILINGS, SIMPLY REPLY TO THIS MESSAGE AND PUT "REMOVE" IN THE SUBJECT.

99 MILLION EMAIL ADDRESSES FOR ONLY $99

Ok, Iknow this is blatantly OT but I'm beginning to go insane. Had an old Dell Dimension XPS sitting in the corner and decided to put it to use, I know it was working pre being stuck in the corner, but when I plugged it in, hit the power nothing happened.

# Bag of words illustration

US Presidential Speeches Tag Cloud
http://chir.ag/projects/preztags/

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# Why naïve Bayes is "naïve"

We call this model "naïve Bayes" because the words aren't ***really*** conditionally independent given the label. For example, the sequence "for you" is more common in spam emails than it would be if the words "for" and "you" were conditionally independent.

**<u>True Statement:</u>**

$$P(X = \text{for you}|Y = \text{Spam}) > P(W = \text{for}|Y = \text{Spam})P(W = \text{you}|Y = \text{Spam})$$

The naïve Bayes approximation simply says: estimating the likelihood of every word sequence is too hard, so for computational reasons, we'll pretend that sequence probability doesn't matter.

**<u>Naïve Bayes Approximation:</u>**

$$P(X = \text{for you}|Y = \text{Spam}) \approx P(W = \text{for}|Y = \text{Spam})P(W = \text{you}|Y = \text{Spam})$$

We use naïve Bayes a lot because, even though we know it's wrong, it gives us computationally efficient algorithms that work remarkably well in practice.

# MPE = MAP using naïve Bayes

Using naïve Bayes, the MPE decision rule is:

$$f(x) = \underset{y}{\operatorname{argmax}}\, P(Y = y) \prod_{i=1}^{n} P(W = w_i | Y = y)$$

P(Y=spam,X=x)=(0.2)(0.01)(0.02)(0.03)(0.03)(0.01) = 36*10^-11

P(Y=h

# Quiz!

- Go to the course web page, click on “27-Jan” to access the 27-Jan quiz on PrairieLearn

# Floating-point underflow

$$f(x) = \underset{y}{\operatorname{argmax}}\, P(Y = y) \prod_{i=1}^{n} P(W = w_i | Y = y)$$

- That equation has a computational issue. Suppose that the probability of any given word is roughly $P(W = w_i | Y = y) \approx 10^{-3}$, and suppose that there are 103 words in an email. Then $\prod_{i=1}^{n} P(W = w_i | Y = y) = 10^{-309}$, which gets rounded off to zero. This phenomenon is called "floating-point underflow."
- In order to avoid floating-point underflow, we can take the logarithm of the equation above:

$$f(x) = \underset{y}{\operatorname{argmax}} \left( \ln P(Y = y) + \sum_{i=1}^{n} \ln P(W = w_i | Y = y) \right)$$

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# Reducing the naivety of naïve Bayes

Remember that the bag-of-words model is unable to represent this fact:

**<u>True Statement:</u>**

$$P(X = \text{for you} | Y = \text{Spam}) > P(W = \text{for} | Y = \text{Spam}) P(W = \text{you} | Y = \text{Spam})$$

Though the bag-of-words model can't represent that fact, we can represent it using a slightly more sophisticated naïve Bayes model, called a "bigram" model.

# N-Grams

Claude Shannon, in his 1948 book *A Mathematical Theory of Communication*, proposed that the probability of a sequence of words could be modeled using N-grams: sequences of N consecutive words.

- **Unigram**: a unigram (1-gram) is an isolated word, e.g., “you”
- **Bigram**: a bigram (2-gram) is a pair of words, e.g., “for you”
- **Trigram**: a trigram (3-gram) is a triplet of words, e.g., “prescription for you”
- **4-gram**: a 4-gram is a 4-tuple of words, e.g., “approved prescription for you”

# Bigram naïve Bayes

A bigram naïve Bayes model approximates the bigrams as conditionally independent, instead of the unigrams. For example,

$$P(X = \text{“approved prescription for you”} | Y = \text{Spam}) \approx$$

$$P(B = \text{“approved prescription”} | Y = \text{Spam}) \times$$
$$P(B = \text{“prescription for”} | Y = \text{Spam}) \times$$
$$P(B = \text{“for you”} | Y = \text{Spam})$$

# Advantages and disadvantages of bigram models relative to unigram models

- Advantage: the bigram model can tell you if a particular bigram is much more frequent in spam than in ham emails.
- Disadvantage: over-training. Even if probabilities of individual words in the training and test corpora are similar, probabilities of bigrams might be different.

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# What are “parameters”?

- Oxford English dictionary: parameter (noun): a numerical or other measurable factor forming one of a set that defines a system or sets the conditions of its operation.
- The naïve Bayes model has two types of parameters:
  - The *a priori* parameters: $P(Y = y)$
  - The *likelihood* parameters: $P(W = w_i | Y = y)$

- In order to create a naïve Bayes classifiers, we must somehow estimate the numerical values of those parameters.

# Parameter estimation

Model parameters: feature likelihoods P(Word | Class) and priors P(Class)

- How do we obtain the values of these parameters?

prior

| | |
|---|---|
| spam: | 0.33 |
| ¬spam: | 0.67 |

P(word | spam)

| | |
|---|---|
| the : | 0.0156 |
| to : | 0.0153 |
| and : | 0.0115 |
| of : | 0.0095 |
| you : | 0.0093 |
| a : | 0.0086 |
| with: | 0.0080 |
| from: | 0.0075 |
| ... | |

P(word | ham)

| | |
|---|---|
| the : | 0.0210 |
| to : | 0.0133 |
| of : | 0.0119 |
| 2002: | 0.0110 |
| with: | 0.0108 |
| from: | 0.0107 |
| and : | 0.0105 |
| a : | 0.0100 |
| ... | |

# Parameter estimation: Prior

The prior, $P(Y)$, is usually estimated in one of two ways.

- If we believe that the test corpus is like the training corpus, then we just use frequencies in the training corpus:

$$P(Y = \text{Spam}) = \frac{\text{Docs}(Y = \text{Spam})}{\text{Docs}(Y = \text{Spam}) + \text{Docs}(Y \neq \text{Spam})}$$

where "Docs(Y=Spam)" means the number of documents in the training corpus that have the label Y=Spam.

- If we believe that the test corpus is different from the training corpus, then we set $P(Y = \text{Spam})$ = the frequency with which we believe spam will occur in the test corpus.

# Parameter estimation: Likelihood

The likelihood, $P(W = w_i | Y = y)$, is also estimated by counting.

The "maximum likelihood estimate of the likelihood parameter" is the most intuitively obvious estimate:

$$P(W = w_i | Y = \text{Spam}) = \frac{\text{Count}(W = w_i, Y = \text{Spam})}{\text{Count}(Y = \text{Spam})}$$

where "$\text{Count}(W = w_i, Y = \text{Spam})$" means the number of times that the word $w_i$ occurs in the Spam portion of the training corpus, and "$\text{Count}(Y = \text{Spam})$" is the total number of words in the Spam portion.

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# What is the probability that the sun will fail to rise tomorrow?

- # times we have observed the sun to rise = 1,825,000
- # times we have observed the sun not to rise = 0
- Estimated probability the sun will not rise = $\frac{0}{0+1,825,000} = 0$

Oops….

# Laplace Smoothing

- The basic idea: add $k$ "unobserved observations" to every possible event
- # times the sun has risen or might have ever risen = 1,825,000+k
- # times the sun has failed to rise or might have ever failed to rise = 0+k
- Estimated probability the sun will rise tomorrow = $\frac{1{,}825{,}000+k}{1{,}825{,}000+2k}$
- Estimated probability the sun will not rise = $\frac{k}{1{,}825{,}000+2k}$
- Notice that, if you add these two probabilities together, you get 1.0.

# Laplace Smoothing for Naïve Bayes

- The basic idea: add $k$ "unobserved observations" to the count of every unigram
  - If a word occurs 2000 times in the training data, Count = 2000+k
  - If a word occur once in training data, Count = 1+k
  - If a word never occurs in the training data, then it gets a pseudo-Count of k
- Estimated probability of a word that occurred Count(w) times in the training data: =

$$P(W = w) = \frac{k + \text{Count}(W = w)}{k + \sum_v (k + \text{Count}(W = v))}$$

- Estimated probability of a word that never occurred in the training data (an "out of vocabulary" or OOV word):

$$P(W = OOV) = \frac{k}{k + \sum_v (k + \text{Count}(W = v))}$$

- Notice that

$$P(W = OOV) + \sum_w \text{P}(W = w) = 1$$

# Conclusions

- MPE = MAP with Bayes' rule:

$$f(x) = \operatorname{argmax}(\log P(Y = y) + \log P(X = x|Y = y))$$

- naïve Bayes:

$$\log P(X = x|Y = y) \approx \sum_{i=1}^{n} \log P(W = w_i|Y = y)$$

- maximum likelihood parameter estimation:

$$P(W = w_i) = \frac{\text{Count}(W = w_i)}{\sum_v \text{Count}(W = v)}$$

- Laplace Smoothing:

$$P(W = w_i) = \frac{k + \text{Count}(W = w_i)}{k + \sum_v (k + \text{Count}(W = v))}$$